

Naina Sharma

DS 4002

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Do Clinical Details Improve Model Diagnosis of Breast Ultrasound Images?

Machine learning is increasingly used in medical imaging to support diagnostic decision-making. However, clinical decisions are rarely based on images alone. In practice, radiologists also consider patient-specific information such as age, BI-RADS score, and lesion characteristics. This case study asks whether incorporating this additional structured information leads to more accurate and reliable predictions of breast cancer lesion images.

In this case study, you will explore how different types of information contribute to the performance of a machine learning model used for breast lesion classification. Specifically, you will investigate whether adding clinical metadata improves predictive accuracy compared to using ultrasound image-derived features alone. You will construct and compare two modeling approaches: an image-only model (a classification model trained using features derived from ultrasound images), and an image + metadata model (incorporates both image-derived features and clinical metadata).

The deliverable: You will use the Breast-Lesions-USG dataset [1], which includes: 256 ultrasound images of breast tissue and lesion masks, and a dataset of clinical data of each patient, including things like patient age, BI-RADS score, and lesion morphology descriptors. Starter scripts have been provided in a GitHub repository to begin this assignment. You will submit your own GitHub repository containing your final code for both models, and generate figures comparing both models to each other.

GitHub Repository Link: <https://github.com/nainasharma22/DS4002>

Reference

[1] The, “A Curated Benchmark Dataset for Ultrasound-Based Breast Lesion Analysis,” The Cancer Imaging Archive (TCIA), Oct. 15, 2024.

<https://www.cancerimagingarchive.net/collection/breast-lesions-usg/>